RiskPulse AI

Explainable financial risk intelligence and portfolio scenario analysis

Trace an event to the positions that carry exposure

Local FinBERT / Multi-source fusion / Financial transmission paths

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S&P Global & Crisil Campus Hackathon 2026

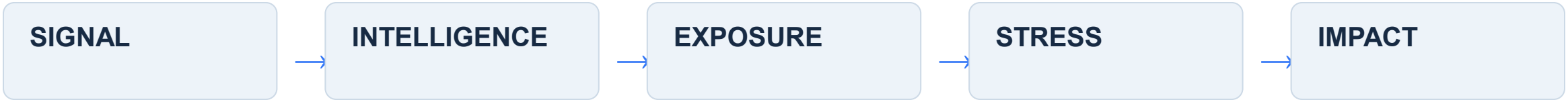
The Problem

Headlines arrive without balance-sheet context

Repeated reports can inflate apparent event activity.
Sentiment alone does not identify exposed positions.
A loss total is difficult to challenge without its assumptions.

The analyst needs an auditable path from evidence
to exposure, scenario loss and the largest contributors.

From Signals to Portfolio Risk



Evidence first

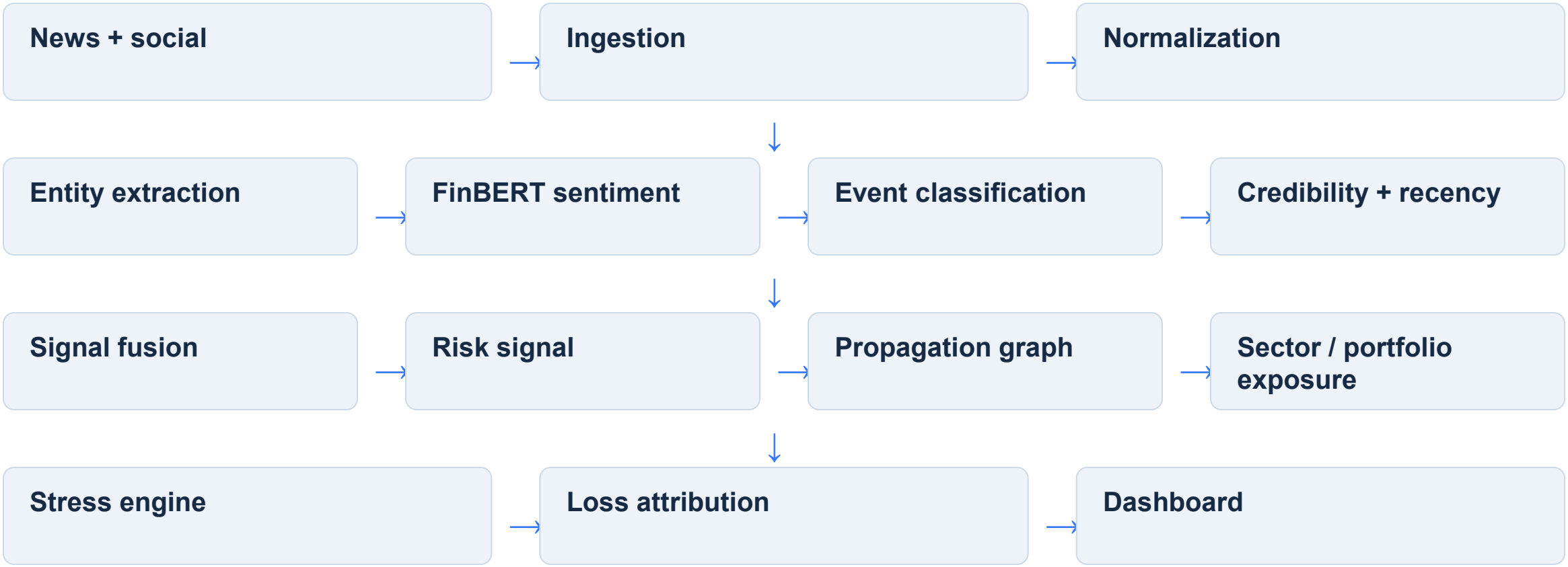
24 observations form 12 event clusters.
Normalized copies add no corroboration.
Source credibility and recency affect priority.
Severity stays separate from confidence.

Portfolio reasoning

One transmission path per position.
Conditional repricing from actual assets.
Sector and position totals reconcile.
Counterfactual inputs are explicit.

Hero: 2 sources, 1 distinct text, Critical effective risk 8.36 / 10

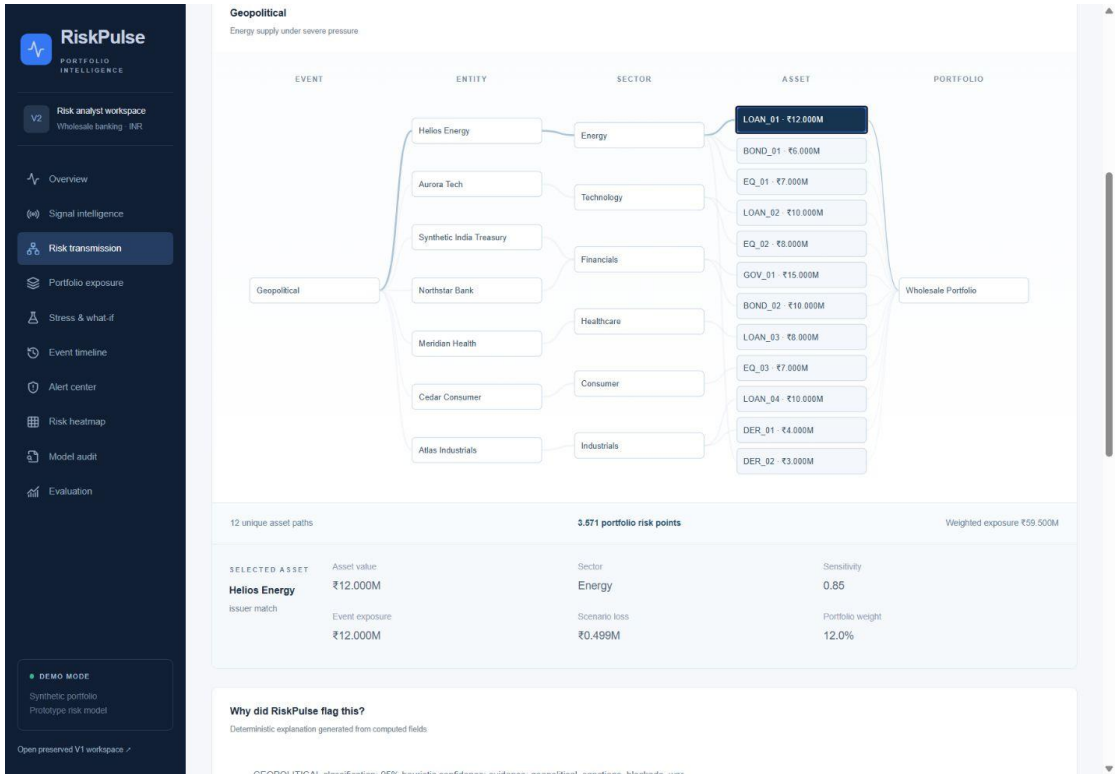
Risk Propagation Architecture



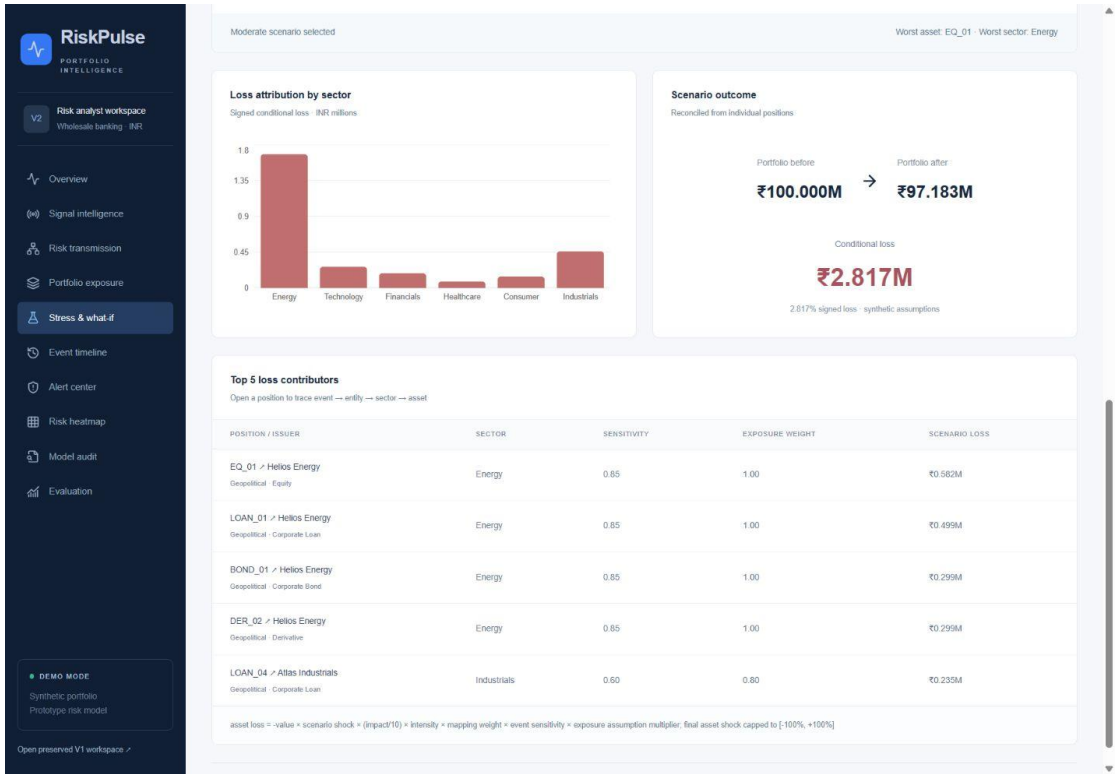
Next.js interface ↔ FastAPI services ↔ SQLite evidence and arrival history

Live Event → Exposure → Stress Test

Event → Risk signal → Propagation → Exposure → Stress → Attribution



12 asset paths / INR 59.5M weighted exposure



INR 100M → 97.183M / Loss INR 2.817M

Actual production application screenshots. Intensity 1.5× gives INR 4.226M loss.

Results + Explainability + Domain Impact

Measured on the local prototype

59 automated tests pass (V1: 25)
24 synthetic records / 12 paired copies
Event-label agreement: 100%
Sentiment agreement: 91.67%

CPU median sentiment inference
Sequential: 24.833 ms per record
Batched: 20.713 ms per record

Traceable financial results

Energy loss: INR 1.679M
Largest asset: EQ_01, INR .582M

Inputs, evidence and edge weights remain available in the audit view.

Analysts can challenge assumptions and compare conditional outcomes.

Small correlated synthetic evaluation. Agreement does not establish real-world accuracy.

Limitations + Future Work

Current limits

Small correlated synthetic dataset
Dictionary entities and lexical similarity
Heuristic confidence and source weights
Illustrative shocks and sensitivities
Simplified instrument repricing
Single-process local application

Validation path

Independent labels and time-based tests
Verified source provenance
Calibrated confidence and market shocks
Hedging and instrument-specific pricing
Bank risk-team review and backtests
Operational security and monitoring

Decision support prototype. No market prediction or regulatory-compliance claim.